Software Requirement Specification (SRS) Document

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Hue&Me — Personal Stylist & Color Analysis Web App

1. Introduction

1.1 Purpose

This document defines the functional, non-functional, and domain requirements for developing Hue&Me, a web application that analyses a user's skin tone and colouring to recommend clothing colours, glasses frames, and makeup shades (foundation, concealer, lipstick, blush) from brands available in the local market. The app also allows users to catalogue their existing wardrobe, build outfits from their own clothes, and preview products on themselves before buying.

This SRS merges the client's requirements with the team's project proposal to form a single, complete, testable specification.

1.2 Scope

Hue&Me will:

- Accept a selfie or a manually entered skin shade (hex code / swatch selection).
- Determine the user's colour type (seasonal palette: Spring / Summer / Autumn / Winter) and undertone (warm/ cool/ neutral/ olive).
- Recommend colours to wear and colours to avoid.
- Recommend glasses frame shapes suited to the user's face shape.
- Match foundation, concealer, lipstick, and blush shades from brands people locally buy.
- Let returning users view their previous analysis results.
- Allow users to select clothing items from the app's database, add them to a personal wardrobe, and view combinations of those items together.

- Provide a static preview of lipstick, glasses, and outfits on the user's uploaded photo.
- Provide an admin interface to maintain the product and shade database.
- Protect user privacy: no selfie storage, no data sharing, full data wipe on request.
- Support accessibility: font size, theme toggle, colour-blind modes, high contrast, text labels alongside swatches.

Target audience: General consumers (primarily in South Asia), beauty shoppers, and the app administrator.

1.3 Definitions, Acronyms, and Abbreviations

- AI: Artificial Intelligence
- CV: Computer Vision
- LAB: CIELAB colour space (perceptually uniform colour model)
- ITA°: Individual Typology Angle (measure of skin depth)
- ΔE: Delta E (colour difference metric)
- AR: Augmented Reality
- API: Application Programming Interface
- PWA: Progressive Web App
- SRS: Software Requirement Specification
- WCAG: Web Content Accessibility Guidelines
- GDPR: General Data Protection Regulation
- JWT: JSON Web Token
- FR: Functional Requirement
- NFR: Non-Functional Requirement

2. System Overview

Hue&Me is a web application built with a React/Next.js frontend and a FastAPI (Python) backend, backed by a PostgreSQL database. It uses MediaPipe (pre-trained computer vision library) for face detection and landmark extraction, and OpenCV for colour-space conversion (RGB → LAB), ITA° calculation, and hue-shift previews. Product shade matching is performed using ΔE colour difference against a curated shade database. No generative AI, no LLM, and no diffusion models are used. All processing is rule based and deterministic.

The system runs on free tier hosting (Vercel + Railway) and is designed to operate at near-zero running cost.

3. Functional Requirements

This section defines the functional capabilities that Hue&Me shall provide. Each requirement is described with inputs, outputs, rationale, priority, and dependencies where applicable.

3.1 User Authentication & Profile

FR1: User Registration

- Description: The system shall allow users to create an account using email and password.
- Inputs: Name, email, password.
- Outputs: Account created, JWT token issued.
- Rationale: Enables personalised history and wardrobe features.
- Priority: High
- Dependencies: Auth service (Firebase Auth or JWT).

FR2: Login / Logout

- Description: The system shall allow registered users to log in and log out securely.
- Inputs: Email, password.
- Outputs: Authenticated session.
- Priority: High

FR3: Password Recovery

- Description: The system shall allow users to reset a forgotten password via email.
- Inputs: Registered email.
- Outputs: Password reset link sent.
- Priority: Medium

FR4: Profile & Preferences

- Description: The system shall store the user's saved colour type, accessibility preferences (font size, theme, colour-blind mode), and language.
- Inputs: User selections.
- Outputs: Persisted preferences applied across sessions.
- Priority: High

FR5: Account Deletion & Data Wipe

- Description: The system shall allow users to delete their account and all associated personal data permanently.
- Inputs: User deletion request.
- Outputs: All user data, including account details, analysis history, wardrobe items, uploaded clothing images, preferences, and any other personal information, will be permanently removed from the database.
- Priority: High
- Dependencies: Database delete operations across all tables.

3.2 Colour Analysis

FR6: Photo Based Skin Analysis

- Description: The system shall accept a user uploaded selfie and determine the user's skin tone, undertone, and seasonal colour type.
- Inputs: Image file (JPG/PNG).
- Outputs: Undertone (warm/cool/neutral/olive), depth (light to medium/tan/deep), ITA°, seasonal palette.
- Rationale: Provides the core analysis driving all recommendations.
- Priority: High
- Dependencies: MediaPipe (Face Detection + Face Mesh), OpenCV.

FR7: Manual Skin Shade Entry

- Description: The system shall allow users to enter their skin shade manually as a hex code or select from a swatch grid, without uploading a photo.
- Inputs: Hex code or swatch selection.
- Outputs: Same analysis output as FR6.
- Rationale: Supports privacy conscious users and users without a suitable photo.
- Priority: High

FR8: Colour Type Result

- Description: The system shall display the user's seasonal colour type (Spring / Summer / Autumn / Winter) with an explanation of how it was derived.
- Inputs: Analysis results from FR6 or FR7.
- Outputs: Seasonal type, detected undertone, detected depth, confidence indicator.
- Priority: High

FR9: Manual Correction

- Description: The system shall allow users to correct the detected tone if they believe it is inaccurate.
- Inputs: User adjusted undertone/depth.
- Outputs: Recalculated palette and recommendations.
- Rationale: Mitigates lighting-based errors and builds user trust.
- Priority: Medium

3.3 Colour Recommendations

FR10: Wear / Avoid Colours

- Description: The system shall list colours the user should wear and colours to avoid, based on their seasonal palette.
- Inputs: Seasonal palette.
- Outputs: Lists of recommended and non-recommended colours with text and hex labels.
- Priority: High

FR11: Glasses Frame Recommendations

- Description: The system shall recommend glasses frame shapes and colours suited to the user's face shape and colour type.
- Inputs: Face shape (detected or manually selected), colour type.
- Outputs: Recommended frame shapes (e.g., rectangular, oval) with rationale.
- Priority: Medium

FR12: Analysis History

- Description: The system shall allow returning users to view their previous analysis results.
- Inputs: Authenticated user session.

- Outputs: List of past analyses with dates and results.
- Priority: High

3.4 Product Shade Matching

FR13: Foundation Matching

- Description: The system shall match the user's skin tone to foundation shades across supported brands using ΔE colour difference.
- Inputs: User LAB skin vector.
- Outputs: Ranked list of foundation shades with match score.
- Priority: High
- Dependencies: Product shade database.

FR14: Concealer Matching

- Description: The system shall match the user's skin tone to concealer shades.
- Inputs: User LAB skin vector.
- Outputs: Ranked list of concealer shades.
- Priority: High

FR15: Lipstick and Blush Matching

- Description: The system shall recommend lipstick and blush shades based on the user's seasonal palette and undertone.
- Inputs: Seasonal palette, undertone.
- Outputs: Ranked list of lipstick and blush shades.
- Priority: High

FR16: Brand Coverage

- Description: The system shall include products from brands commonly purchased in the local market (e.g., Maybelline, L'Oréal, etc)
- Inputs: Admin-maintained product list.
- Outputs: Product recommendations drawn from this list.
- Priority: High
- Dependencies: Admin product management (FR23).

FR17: Product Links

- Description: Each recommended product shall include a link to the brand's website or a retailer page (affiliate links are future work).
- Inputs: Product record.
- Outputs: Clickable link.
- Priority: Low

3.5 Wardrobe & Outfit Builder

FR18: Add Clothing Items

- Description: The system shall allow users to browse a database of clothing items (maintained by the admin) and add selected items to their personal wardrobe.
- Inputs: Selection of clothing item(s) from the app's database.
- Outputs: Selected items saved to the user's wardrobe.
- Rationale: Client requested a simple wardrobe where users can collect items from the app's catalogue.
- Priority: High
- Dependencies: Admin-managed clothing database.
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FR19: Colour Compatibility Check

- Description: The system shall indicate whether each wardrobe item is compatible with the user's colour type.
- Inputs: Wardrobe item colour, user's seasonal palette.
- Outputs: Compatible / not compatible flag with explanation.
- Priority: High

FR20: Outfit Builder

- Description: The system shall allow users to select multiple items from their wardrobe and view them together as a visual combination.
- Inputs: Selected wardrobe items.
- Outputs: Visual collage of selected items (e.g., top + bottom + accessory).
- Priority:Medium

3.6 Preview (Static Try-On)

FR21: Static Product Preview

- Description: The system shall allow users to preview a lipstick shade, glasses frame, or outfit on their uploaded photo as a static overlay (not real-time AR).
- Inputs: User photo, selected product/outfit.
- Outputs: Still image with the product overlaid.
- Priority: High
- Note: Real-time AR try-on is out of scope (see Section 7).

FR22: Preview Performance

- Description: The static preview shall render within a few seconds of product selection.
- Inputs: Selected product + user photo.
- Outputs: Rendered preview image.
- Priority: Medium

3.7 Admin Functions

FR23: Product Database Management

- Description: The system shall provide an admin interface to add, edit, and remove products and shades.
- Inputs: Admin credentials, product data (brand, name, category, shade, hex, LAB values).
- Outputs: Updated product database.
- Priority: High

FR24: User Management

- Description: The system shall allow the admin to view and manage user accounts (e.g., deactivate, delete on request).
- Inputs: Admin credentials.
- Outputs: Updated user records.
- Priority: Medium

3.8 Accessibility

FR25: Font Size Control

- Description: The system shall allow users to increase or decrease the app's font size.

- Inputs: User preference.
- Outputs: Applied font size across the app.
- Priority: High

FR26: Theme Toggle (Day / Night)

- Description: The system shall provide a light mode and dark mode.
- Inputs: User preference.
- Outputs: Applied theme across the app.
- Priority: High

FR27: Colour-Blind Support

- Description: The system shall not rely on colour swatches alone. Every colour recommendation shall include a text label (e.g., "Rust — warm orange-brown") and a hex code. Colour-blind modes (protanopia, deuteranopia, tritanopia) shall be available.
- Inputs: User preference.
- Outputs: Adjusted colour display + text labels.
- Priority: High

FR28: High Contrast Option

- Description: The system shall offer a high-contrast mode for users with poor eyesight.
- Inputs: User preference.
- Outputs: High-contrast UI.
- Priority: Medium

3.9 Privacy

FR29: No Photo Storage

- Description: The system shall not store user-uploaded selfies on the server after analysis is complete.
- Inputs: Uploaded image.
- Outputs: Image processed in memory and discarded.
- Priority: High

FR30: No Data Sharing

- Description: The system shall not share any personal user data with third parties.
- Inputs: User data.
- Outputs: No external transmission of personal data.
- Priority: High

4. Non-Functional Requirements

4.1 Performance

- NFR1: The system shall return colour analysis results within a few seconds of image upload or manual entry.
- NFR2: The static preview (FR21) shall render within a few seconds of product selection.
- NFR3: The system shall support at least 100 concurrent users on free-tier hosting.

4.2 Reliability

- NFR4: The system shall have an uptime of 99% (constrained by free-tier hosting).
- NFR5: The system shall retry failed API calls up to 3 times before showing an error.

4.3 Security

- NFR6: All communications shall use HTTPS/SSL.
- NFR7: Passwords shall be hashed using bcrypt or equivalent.
- NFR8: Authentication shall use JWT tokens with expiry.
- NFR9: Uploaded images shall be processed in memory and not written to disk.

4.4 Privacy & Compliance

- NFR10: The system shall comply with GDPR principles for data minimisation and right to erasure.
- NFR11: Users shall be able to delete their account and all associated data at any time.

4.5 Accessibility

- NFR12: The system shall meet WCAG 2.1 Level A for colour contrast and text alternatives. Full AA compliance is future work.
- NFR13: The system shall function without relying on colour alone to convey information.

4.6 Scalability

- NFR14: The system shall handle increased load during peak usage without crashing (within free-tier limits).

4.7 Cost

- NFR15: The system shall operate at $0/month using free-tier hosting (Vercel, Railway) and free product data sources (Open Beauty Facts, Makeup API) or a manually curated shade table.

4.8 Mobile Responsiveness

- NFR16: The system shall be fully responsive and usable on mobile devices (phones and tablets).

5. Interface Requirements

5.1 User Interface (UI)

The system shall be accessible on:

- Desktop web browsers (Chrome, Firefox, Safari, Edge).
- Mobile web browsers (responsive design).
- Installable as a PWA (Progressive Web App) on mobile devices.

5.2 APIs

The system shall connect with:

- MediaPipe (library, runs locally — not an external API).
- OpenCV (library, runs locally).
- Open Beauty Facts API (free, no auth) — for product data.

- Makeup API (free, no auth) — for product shade data.
- PostgreSQL — internal database.

5.3 Hardware Interfaces

- The system shall be deployed on Vercel (frontend) and Railway (backend + database) free tiers.
- No GPU, no specialised hardware required.

6. Domain Requirements

- DR1: The system shall use CIELAB colour space and ITA° for skin tone classification, following established colour science standards.
- DR2: Product shade matching shall use ΔE (Delta E) colour difference, with a threshold of $\Delta E \leq 3.0$ for a "close match."
- DR3: The system shall comply with local consumer protection laws regarding product recommendations (no misleading claims).
- DR4: The system shall not claim medical or dermatological accuracy — it is a styling tool, not a diagnostic tool.

7. Constraints

- Development shall be completed within 7 weeks (2 sprints).
- The system shall operate on free-tier hosting only (no paid services).
- Real-time AR try-on is out of scope, only static preview is included.
- Generative AI / diffusion models are out of scope.
- LLM-based text generation is out of scope — templates only.
- Custom AI model training is out of scope — MediaPipe is pre-trained.
- E-commerce checkout / payment processing is out of scope.
- Native mobile apps (iOS/Android) are out of scope — web-first, PWA optional.
- Brand partnership contracts / commission tracking are out of scope.

8. Assumptions and Dependencies

- Users will have a device with a camera or access to a photo of themselves.
- Users will have a stable internet connection.
- MediaPipe will continue to be available as a free, open-source library.

- Open Beauty Facts / Makeup API will remain accessible (fallback: manually curated shade table).
- The admin will maintain the product database after launch.
- Free-tier hosting limits (Vercel, Railway) will be sufficient for the demo scale.

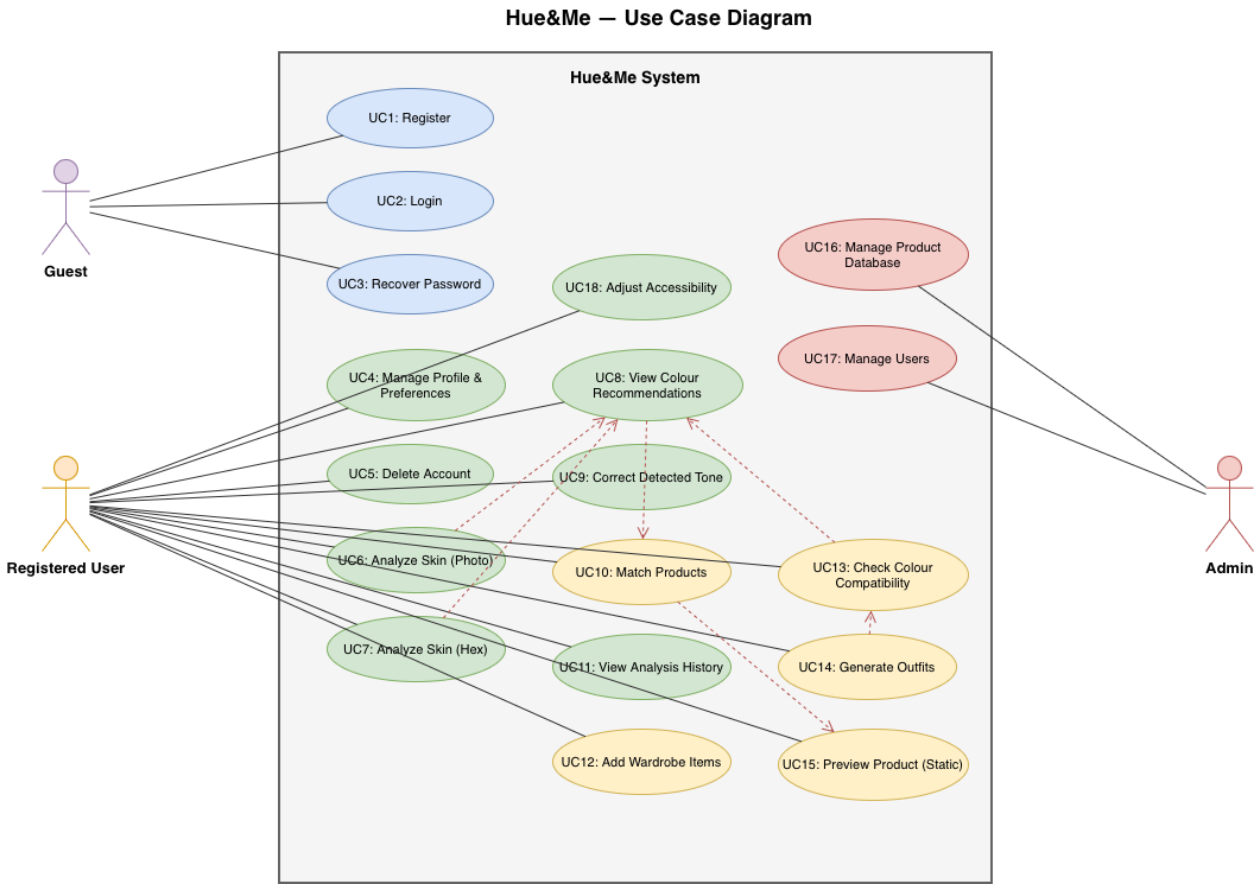
9. SDLC Methodology

- Agile — Scrum
- 2 Sprints (2 core features each)
- Sprint Planning, Daily Scrums, Sprint Review, Sprint Retrospective
- Product Backlog maintained on Trello

10. Appendices

- A1: Glossary of technical terms (LAB, ITA°, ΔE, MediaPipe, PWA).
- A2: Links to relevant standards:
    - WCAG 2.1 Guidelines
    - GDPR Guidelines
    - CIELAB Colour Space Reference
    - MediaPipe Documentation

Part 1: Use Case Diagram



Part 2: Use Case Descriptions

UC1: Register

Field	Description
Actor	Guest
Precondition	User has a valid email address
Trigger	User clicks "Sign Up"

Main Flow	1. User enters name, email, password. 2. System validates input. 3. System checks email uniqueness, 4. System hashes password (bcrypt). 5. The system creates an account. 6. System issues JWT token. 7. System redirects to dashboard
Alternate Flow	1. Email already exists → show error 2a. Invalid email format → show error b. Weak password → show requirements
Postcondition	User account created, user logged in
Priority	High

UC2: Login

Field	**Description**
Actor	Guest
Precondition	User has a registered account
Trigger	User clicks "Login"
Main Flow	1. User enters email + password. 2. System validates credentials 3. System issues JWT token 4. System redirects to dashboard
Alternate Flow	2a. Invalid credentials → show error 2b. Account deactivated → show message
Postcondition	User authenticated, session active

Priority	High

UC3: Recover Password

Field	**Description**
Actor	Guest
Precondition	User has a registered email
Trigger	User clicks "Forgot Password"
Main Flow	1. User enters email 2. System verifies email exists 3. System sends reset link 4. User clicks link, enters new password 5. System updates password hash
Alternate Flow	2a. Email not found → show generic message (security)
Postcondition	Password updated
Priority	Medium

UC4: Manage Profile & Preferences

Field	**Description**
Actor	Registered User
Precondition	User is logged in
Trigger	User opens Settings

Main Flow	1. User views current preferences 2. User changes font size / theme / colour-blind mode / language 3. System saves preferences 4. System applies changes across app
Postcondition	Preferences persisted
Priority	High

UC5: Delete Account (& Data)

Field	**Description**
Actor	Registered User
Precondition	User is logged in
Trigger	User clicks "Delete Account"
Main Flow	1. System shows confirmation dialog 2. User confirms 3. System deletes all user data (analyses, wardrobe, preferences, matches) 4. System deletes account 5. System redirects to home
Alternate Flow	2a. User cancels → no action
Postcondition	All user data permanently removed (GDPR compliance)
Priority	High

UC6: Analyze Skin (Photo)

Field	Description
Actor	Registered User
Precondition	User is logged in
Trigger	User uploads selfie
Main Flow	1. User uploads JPG/PNG selfie 2. System processes image in memory (not stored) 3. MediaPipe detects faces + 468 landmarks 4. The system extracts cheek + forehead pixels 5. The system converts RGB → LAB 6. The system calculates ITA° 7. System classifies undertone + depth 8. System maps to seasonal palette 9. System displays results
Alternate Flow	1. No face detected → show error, suggest retry 2. Multiple faces → ask the user to select 3. Invalid file format → show error
Postcondition	Analysis saved, image discarded
Priority	High
Includes	UC8 (View Color Recommendations)

UC7: Analyze Skin (Hex/(Swatch)

Field	Description
Actor	Registered User
Precondition	User is logged in

Trigger	User selects "Enter manually"
Main Flow	1. User enters hex code OR selects swatch 2. System validates hex format 3. System converts hex → RGB → LAB 4. System calculates ITA° 5. System classifies undertone + depth 6. System maps to seasonal palette 7. System displays results
Alternate Flow	2a. Invalid hex → show error
Postcondition	Analysis saved (no image processed)
Priority	High
Includes	UC8 (View Color Recommendations)

UC8: View Color Recommendations

Field	**Description**
Actor	Registered User
Precondition	Analysis completed (UC6 or UC7)
Trigger	Analysis results displayed
Main Flow	1. System shows seasonal palette. 2. System lists "wear" colors with hex + text labels. 3. System lists "avoid" colors with hex + text labels. 4. System shows glasses frame recommendations 5. System explains how derived
Postcondition	User sees recommendations

Priority	High
Includes	UC10 (Match Products)

UC9: Correct Detected Tone

Field	**Description**
Actor	Registered User
Precondition	Analysis completed
Trigger	User clicks "This doesn't look right"
Main Flow	1. System shows adjustment controls 2. User adjusts undertone/depth 3. System recalculates palette 4. System updates recommendations 5. System marks analysis as corrected
Postcondition	Analysis updated
Priority	Medium

UC10: Match Products

Field	**Description**
Actor	Registered User
Precondition	Analysis completed (UC 8)
Trigger	User views recommendations

Main Flow	1. System reads user LAB vector. 2. System queries product shade database. 3. System calculates ΔE for each shade. 4. System filters ΔE ≤ 3.0 5. System ranks by ΔE ascending. 6. System displays matches with scores. 7. Each match includes buy link
Alternate Flow	2a. Database empty → show message. 2b. No matches within threshold → show closest
Postcondition	Product matches displayed and saved
Priority	High
Include	UC15 (Preview Product)

UC11: View Analysis History

Field	**Description**
Actor	Registered User
Precondition	User has previous analyses
Trigger	User opens History
Main Flow	1. System retrieves past analyses. 2. System displays list with dates + results. 3. User clicks an analysis to view details
Postcondition	History displayed
Priority	High

UC12: Add Wardrobe Items

Field	Description
Actor	Registered User
Precondition	User is logged in
Trigger	User clicks "Add Item"
Main Flow	1. a) The user browses the clothing database. b) The user selects one or more items. 2. System adds selected items to the user's wardrobe. 3. The system checks compatibility with the user's palette.
Postcondition	Selected items saved to the wardrobe.
Priority	High

UC13: Check Color Compatibility

Field	Description
Actor	Registered User
Precondition	Wardrobe item exists
Trigger	User views wardrobe
Main Flow	1. System reads item colour. 2. System compares with user's palette. 3. System flags compatible / not compatible. 4. System shows explanation
Postcondition	Compatibility displayed

Priority	High
Includes	UC8 (View Color Recommendations)

UC14: Generate Outfits

Field	**Description**
Actor	Registered User
Precondition	User has ≥ 2 wardrobe items
Trigger	User clicks "Generate Outfits"
Main Flow	1. The user selects multiple items from their wardrobe. 2. The system displays selected items together as a visual collage. 3. Users can save the combination if desired.
Alternate Flow	2a. Not enough items → show message
Postcondition	Visual combination displayed.
Priority	Medium
Includes	UC13 (Check Color Compatibility)

UC15: Preview Product (Static)

Field	**Description**
Actor	Registered User
Precondition	User has uploaded photo + selected product

Trigger	User clicks "Preview"
Main Flow	1. System retrieves user photo. 2. System detects face landmarks. 3. System overlays product (lipstick / glasses / outfit) as a static image. 4. System renders promptly
Alternate Flow	2a. No face detected → show error
Postcondition	Static preview displayed
Priority	High
Note	Real-time AR is out of scope

UC16: Manage Product Database

Field	**Description**
Actor	Admin
Precondition	Admin is logged in
Trigger	Admin opens admin panel
Main Flow	1. Admin views product list. 2. Admin adds / edits / removes products. 3. Admin adds shades with hex + LAB values. 4. System updates database
Postcondition	Product database updated
Priority	High

UC17: Manage Users

Field	Description
Actor	Admin
Precondition	Admin is logged in
Trigger	Admin opens user management
Main Flow	1. Admin views user list. 2. Admin can deactivate / delete accounts. 3. System updates user records
Postcondition	User records updated
Priority	Medium

UC18: Adjust Accessibility

Field	Description
Actor	Registered User
Precondition	User is logged in
Trigger	User opens accessibility settings
Main Flow	1. User adjusts font size. 2. User toggles theme (day/night). 3. User selects colour-blind mode. 4. User enables high contrast. 5. System applies changes immediately
Postcondition	Accessibility preferences saved
Priority	High

Part 2: User Stories (15 Stories)

Sprint 1 Stories

ID	Title	User Story	Acceptance Criteria	Priority
US1	User Registration	As a guest, I want to register with email and password so that I can save my analysis history.	1. Valid email required. 2. Password ≥ 8 chars. 3. Email uniqueness checked. 4. JWT issued. 5. Error on duplicate email	High
US2	Login / Logout	As a registered user, I want to log in and log out so that my data stays secure.	1. Valid credentials → JWT. 2. Invalid → error. 3. Logout clears the token. 4. Session persists across refresh.	High
US3	Password Recovery	As a registered user, I want to recover my password so that I can regain access if I forget it.	1. Valid email → reset link sent. 2. The link expires after 1 hour. 3. New password meets requirements	Medium
US4	Manage Preferences	As a registered user, I want to manage my preferences so that the app suits my needs.	1. Font size, theme, colour-blind mode saved. 2. Preferences persist. 3. Changes apply immediately	High
US5	Delete Account& Data	As a registered user, I want to delete my account and all data so that my privacy is protected.	1. Confirmation dialog. 2. All data deleted. 3. Account removed. 4. Redirect to home	High

US6	Skin Tone Analysis (Photo)	As a registered user, I want to upload a selfie so that the app detects my skin tone.	1. JPG/PNG accepted. 2. Face detected via MediaPipe. 3. LAB values extracted. 4. ITA° calculated. 5. Undertone classified. 6. Image discarded	High
US7	Manual Skin Tone Entry	As a registered user, I want to enter my skin tone manually so that I can use the app without a photo.	1. Hex code input accepted. 2. Swatch grid available. 3. Same output as photo analysis. 4. Invalid hex → error	High
US8	View Seasonal Color Type	As a registered user, I want to see my seasonal colour type so that I know which colours suit me.	1. Seasonal palette displayed. 2. Undertone + depth shown. 3. Explanation of derivation. 4. Confidence indicator	High
US9	Wear / Avoid Colors	As a registered user, I want to see colours to wear and avoid so that I can shop confidently.	1. Wear list with hex + text labels. 2. Avoid list with hex + text labels. 3. Accessible (text + swatch)	High
US10	Correct Detected Tone	As a registered user, I want to correct my detected tone so that the app matches my real skin.	1. Adjustment controls shown. 2. Palette recalculated. 3. Analysis marked corrected	Medium

Sprint 2 Stories

ID	Title	User Story	Acceptance Criteria	Priority
US11	Foundation Matching	As a registered user, I want to match foundation shades so that I can buy the right product.	1. ΔE calculated per shade. 2. Ranked by ΔE ascending. 3. ΔE ≤ 3.0 threshold. 4. Match score displayed. 5. Buy link included	High
US12	Concealer Matching	As a registered user, I want to match concealer shades so that I can cover blemishes naturally.	1. Same as US11 for concealers. 2. Shade data from database	High
US13	Lipstick & Blush Recommendations	As a registered user, I want to get lipstick and blush recommendations so that my makeup matches my palette.	1. Based on the seasonal palette. 2. Ranked by suitability. 3. Hex + text labels	High
US14	Add Wardrobe Items	As a registered user, I want to browse a clothing database and add items to my wardrobe so that I can collect pieces I like.	1. User can browse the clothing database 2. User can select one or more items to add 3. Selected items appear in the user's wardrobe..	High
US15	Check Color Compatibility	As a registered user, I want to check if my clothes suit my colours so that I know what to wear.	1. Compatibility flag shown. 2. Explanation provided. 3. Based on seasonal palette	High
US16	View Items Together	As a registered user, I want to view different pieces of clothing together so that I can see how	1. User can select multiple wardrobe items. 2. Selected items are displayed together as a visual collage.	Medium

		they look as a combination.	3. User can save the combination.	
US17	Static Product Preview	As a registered user, I want to preview products on my photo so that I can see how they look.	1. Static overlay (not AR). 2.Lipstick/glasses/outfit. 3. Renders promptly without noticeable delay. 4. Face landmarks used for positioning.	High
US18	View Analysis History	As a registered user, I want to view my past analyses so that I can revisit results.	1. List with dates. 2. Click to view details. 3. Sorted by date descending	High
US19	Accessibility Settings	As a registered user, I want to adjust accessibility settings so that the app is usable for me.	1. Font size control. 2. Theme toggle. 3. Colour-blind modes. 4. High contrast option	High
US20	Admin Product Management	As an admin, I want to manage the product database so that recommendations stay accurate.	1. Add/edit/remove products. 2. Add shades with hex + LAB. 3. Changes reflect immediately.	High

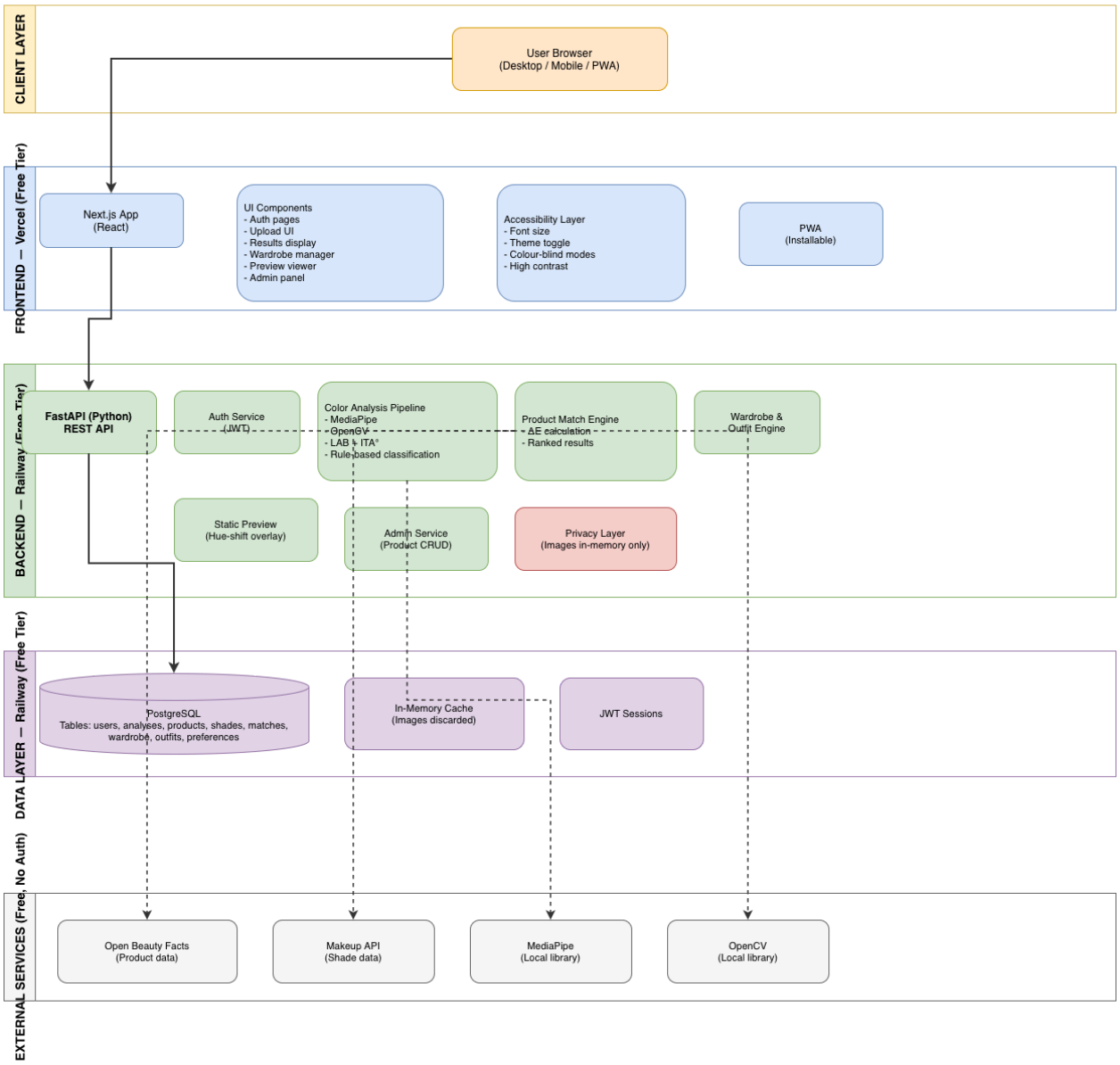
Traceability Matrix (FR → UC → US)

FR	**Use Case**	**User Story**
FR1	UC1	US1
FR2	UC2	US2
FR3	UC3	US3

FR4	UC4	US4
FR5	UC5	US5
FR6	UC6	US6
FR7	UC7	US7
FR8	UC8	US8
FR9	UC9	US10
FR10	UC8	US9
FR11	UC8	—
FR12	UC11	US18
FR13	UC10	US11
FR14	UC10	US12
FR15	UC10	US13
FR16	UC10	—
FR17	UC10	—
FR18	UC12	US14
FR19	UC13	US15
FR20	UC14	US16
FR21	UC15	US17

FR22	UC15	US17
FR23	UC16	US20
FR24	UC17	—
FR25	UC18	US19
FR26	UC18	US19
FR27	UC18	US19
FR28	UC18	US19
FR29	UC6	US6
FR30	UC5	–

Hue&Me — System Architecture



Hue&Me — Entity Relationship Diagram (ERD)

